Transformers for Enhanced Biomedical Text Classification

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***Abstract*- This research delves into advanced methods for sequence classification in biomedical texts, emphasizing the role of deep learning models that leverage hybrid embeddings to enhance both accuracy and interpretability. The work is designed to systematically cover the style in which the token, character, as well as positional embeddings are employed in neural networks. These are Recurrent Neural Network (RNNs), Convolutional Neural Network (CNNs) and other Transformer models including BERT and Self-Attention. A new model based on BERT but enhanced with positional encoding is a valuable addition of this study.**

***Keywords*- Bidirectional Encoder Representations from Transformers, Natural Language Processing, Convolutional Neural Network, Recurrent Neural Network, Long short-term memory.**

I. INTRODUCTION

Due to the large amount of biomedical literature now available, it becomes imperative to have good mechanisms to classify such information as in the cases of information retrieval and text summarization [1]. In traditional models, including RNNs and CNNs, domain-specific context in medical texts usually cannot be identified effectively. Nonetheless, general transformer models such as BERT have yielded high performance and thereby there are models for specific domain such as BioBERT for bio-medical terms [2]. More enhancements of the BERT-based models will continue to offer prospects in the enhancement of biomedical text classification [3]. Many of the prior approaches toward the classification of biomedical text often times fail to address the intricacies and specificity of medical documents. This is made worse by the fact that the texts under discussion employ the medical language comprehensible to professionals only and exhibit different levels of formalization. Hence, it is high time that models that can show the characteristics of biomedical text be refined in order to yield improved classification results.

The objective of this study is to improve the biomedical text classification problem by proposing new models for the fused token, character, and position embeddings. These models are designed to deal with the specifics of the context in which the medical literature is presented, which typically are the use of the medical terminology and the heterogeneity of the formats. In addition to that, the study will also establish the comparative efficiency of the different types of neural network architectures such as the RNNs, CNNs, and Transformer models [4]. Further, accuracy and robustness will be deemed most important in applied research in the biomedical field. The research makes several significant contributions to the field of biomedical text classification, offering insights and advancements in model development and evaluation:

- **Introducing a BERT-based model with positional encoding**: However, here we propose a novel model based on BERT with Positional encoding, F1 scores are slightly worse, but substantially better accuracy on PubMed 200k RCT dataset. From the analysis shown in the paper, we conclude that the proposed model is able to extract contextual and structural features from text of biomedical learning.

- **Comprehensive evaluation of multiple Neural Network architectures**: At the same time, the study also provides a comprehensive analysis of the performance of RNNs, CNNs, Transformer models and Self Attention mechanisms when used with a hybrid model. (TILL HERE)
- **Practical insights for the design and optimization of biomedical text classification models**: The research offers valuable guidelines for designing, optimizing, and applying neural networks in biomedical text classification, with a focus on improving accuracy and robustness in real-world applications like information retrieval and systematic reviews.

II. RELATED WORK

Biomedical text classification has received substantial research focus due to the extensive volume of biomedical literature [5]. This section summarizes major approaches explored in this area and compares them to the techniques used in this study.

A. Traditional machine learning approaches

Early work used algorithms like SVMs, Decision Trees, and Naive Bayes with manually crafted features such as n-grams and TF-IDF vectors. While foundational, these models struggled to capture the deeper semantics of biomedical texts due to their reliance on manual feature extraction [3] [6].

B. Deep learning models

With the rise of deep learning, CNNs and RNNs became widely adopted for text classification. CNNs were useful for short text sequences by identifying local patterns, while RNNs, especially LSTMs, handled longer text sequences by capturing dependencies [1] [7]. However, both have limitations in processing long-range dependencies in complex biomedical texts [4].

C. Transformer models

The Transformer model, presented by Vaswani, transformed the field of NLP [4] [5]. BERT, in particular, brought bidirectional context understanding, improving text classification accuracy [8] [9]. Specialized models like BioBERT and ClinicalBERT further enhanced processing for medical texts [2].

D. Hybrid models in biomedical text classification

Recent studies have explored hybrid models that integrate token and character embeddings to improve performance [8] [10]. Combining CNNs, RNNs, and Transformers has also been effective, although hybrid approaches in biomedical text classification are still relatively underexplored [9] [11].

E. Challenges and limitations in existing work

Despite advancements, challenges remain in handling the specialized vocabulary and structure of medical texts, and deep learning models often lack interpretability [12].

F. Comparison to the present study

This study advances prior work by introducing a BERT-based model with positional encoding, combining token and character embeddings to capture both semantic and structural aspects of biomedical texts. It also provides a comprehensive evaluation of different architectures and hybrid models [3].

III. METHODOLOGY

This section covers the datasets used, preprocessing steps, model architectures, training processes, and evaluation methods to develop and measure the performance of the proposed models for biomedical text classification.

A. Dataset

<u>PubMed 200k RCT Dataset</u>: The study utilized the PubMed 200k Randomized Controlled Trials (RCT) dataset, which consists of around 200,000 abstracts from biomedical literature. The dataset includes over 2.3 million sentences, each labeled under one of five categories: background, objective, method, result, or conclusion. The dataset is ideal for sequence classification tasks due to its size and structured nature [1]. <u>Dataset Splitting</u>: The data was divided into training, validation, and test sets, with 70% assigned to training, 15% to validation, and 15% to testing. This division ensured that the models were trained on ample data while maintaining a robust evaluation process on unseen samples.

B. Data preprocessing

Tokenization and text vectorization: Tokenization was conducted to break the text into tokens (words or subwords). For initial vectorization, Term Frequency-Inverse Document Frequency (TF-IDF) was applied, which converted tokens into numerical representations based on their significance across the entire dataset.

Character-level embeddings: In addition to token embeddings, character-level embeddings were created to capture detailed text features, which are useful in processing medical terminology and abbreviations. This enabled the models to learn from the structure of words, enhancing their ability to differentiate between similar terms [13].

Positional encoding: To account for the structure of sentences, positional encoding was added to the embeddings [13]. This technique encodes the position of each word within a sentence, helping the models better understand the order and context of words, which is crucial for accurate sequence classification.

Label encoding: The text labels (background, objective, method, result, conclusion) were converted into numerical values using label encoding. This conversion allowed the labels to be processed efficiently by machine learning algorithms.

C. Model architectures

Model 1: Conv1D with token embeddings

The first model employed a CNN with 1-dimensional convolutional layers (Conv1D) applied to token embeddings. This model was designed to capture local patterns within text sequences, which is useful for analyzing shorter sequences and identifying n-gram features in the text.

Model 2: Universal Sentence Encoder (USE)

The second model used the Universal Sentence Encoder (USE), a pre-trained model from TensorFlow Hub, to convert entire sentences into high-dimensional vectors. These vectors represent the semantic content of the sentences and were passed through dense layers for classification.

Model 3: Conv1D with character embeddings

This model combined Conv1D layers with character-level embeddings. It aimed to capture intricate details within the text, particularly useful for distinguishing biomedical terminologies, by analyzing the text at a character level.

Model 4: Hybrid token and character embeddings

Model 4 integrated both token and character embeddings and used Conv1D layers. This hybrid approach aimed to take advantage of the strengths of both embedding methods by capturing semantic information from tokens while preserving the detailed structural features of words.

Model 5: Tribrid embeddings with positional features

Model 5 as depicted in figure 1, expands on the hybrid approach by adding positional encoding to token and character embeddings. This tribrid model aimed to better understand the structure and order of the sequence, combining insights from the three different types of embeddings (token, character, and positional). The model 5 begins by processing the text through three distinct input layers:

- **Token inputs:** The text is tokenized and vectorized, converting the raw words into numerical vectors representing the semantic meaning of the words.
- **Character inputs:** A separate vectorization process is applied to characters, capturing finer details such as prefixes, suffixes, and subword patterns.
- **Positional features:** Structural information, such as the position of a sentence within the abstract, is included to provide additional context.

After the embeddings are created, the outputs from the token and character embeddings are concatenated to form a unified representation. This combined embedding is further processed through dense layers to capture the interactions between the features. Finally, the positional embeddings are integrated, providing the model with both content and structural information as shown in figure 1.

The final dense layers then aggregate these rich embeddings, outputting the classification results for each sentence.

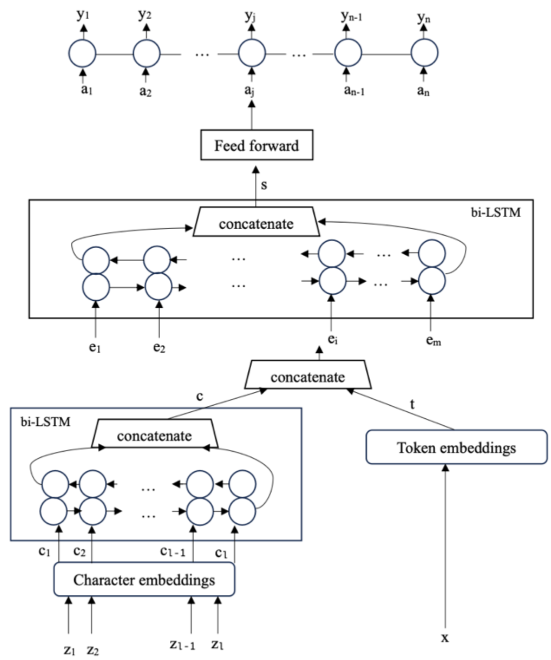

Figure 1: Architecture of the ANN model for sequential sentence classification.

Input tokens x are transformed into token embeddings t (300 dimensions) and character embeddings c_i (25 dimensions) for each character z_i. These are processed through a bi-LSTM to generate character-based token embeddings c(50 dimensions). The token and character embeddings are concatenated to form hybrid token embeddings e_i (350 dimensions). The bi-LSTM then produces a sentence vector s (200 dimensions), which is passed through a feed-forward layer to generate sentence label vectors a_j, corresponding to the predicted sentence labels y_j. Token embeddings are initialized using pretrained glove embeddings on large datasets such as wikipedia and gigaword.

Model 6: BERT with positional encoding

A BERT-based model enhanced with positional encoding was introduced as model 6. Model 6 as depicted in figure 2, utilizes BERT (Bidirectional Encoder Representations from Transformers) as its core, making the most of its ability to capture contextual relationships from both directions within a text [8]. The added positional encoding further helped the model understand the sequential structure of abstracts [14]. Model 6 achieved the highest performance in terms of accuracy and F1-score.

The architecture of Model 6 (figure 2) is structured as:

- **BERT layer:** Input tokens are processed by a pre-trained BERT model, which generates contextual embeddings for every token. BERT's strength comes from its ability to understand the surrounding context of each word, making it particularly effective for classifying sentences [14].
- **Character embeddings:** In addition to the BERT embeddings, character-level embeddings are also generated through a separate layer. These embeddings focus on the detailed structure of the words, such as biomedical terms, complementing the broader contextual understanding from BERT [14].
- **Concatenation and positional encoding:** The contextual embeddings from BERT and the character-level embeddings are concatenated to form a richer representation. Positional encoding is used to ensure the model considers the order of words in a sentence, which is essential for understanding the sequence's structure.
- **Dense layers**: The concatenated embeddings are processed through dense layers that refine the learned features, combining the contextual and positional information to make accurate classifications.
- **Output layer:** Finally, the refined embeddings are passed through the output layer, where the sentences are classified into their respective categories (e.g., background, objective, method, result, or conclusion).

By combining BERT's deep contextual understanding with character-level details and positional information, Model 6 achieves high accuracy and strong F1 scores, making it one of the most robust models in this study.

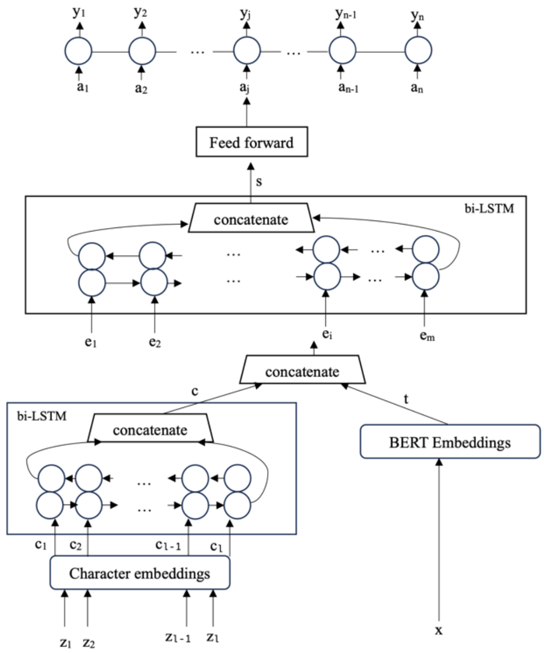

Figure 2: BERT with positional encoding architecture

Similar to Figure 1, but replacing token embeddings with BERT embeddings.

Model 7: Self-attention with Universal Sentence Encoder (USE)

The final model, model 7 as depicted in figure 3, combined a self-attention mechanism with the Universal Sentence Encoder (USE). This model aimed to capture long-range dependencies within the text, with the self-attention mechanism allowing the model to more effectively access the importance of different text parts. It integrates the Universal Sentence Encoder (USE) with a self-attention mechanism. The model 7 (figure 3) is structured as follows:

- **Universal Sentence Encoder (USE):** The model begins with USE, which converts entire sentences into high-dimensional vectors. These vectors capture the overall meaning of the sentences, making it suitable for tasks requiring semantic understanding [7].

- **LSTM layers:** The sentence embeddings from USE are fed into bidirectional LSTM (Long Short-Term Memory) layers, which capture dependencies in both forward and backward directions. This is essential for understanding the context within sequences [7] [5].
- **Additive attention mechanism:** To highlight important parts of the text, an additive attention mechanism is used. This mechanism enables the model to evaluate the relevance of different words, prioritizing the more informative ones for the classification task.
- **Positional and contextual features:** The model also integrates positional and contextual features, such as the sentence position within the abstract. These features are processed through dense layers and combined with the attention-weighted output from the LSTM layers [7].
- **Dense and output layers:** Finally, the combined embeddings are passed through dense layers, refining the representations further before classifying the sentences into their respective categories.

Model 7 excels in situations where understanding long-range dependencies and the overall semantic content of the text is crucial. The combination of USE, LSTM layers, and the self-attention mechanism enables this model to perform effectively in complex text classification tasks.

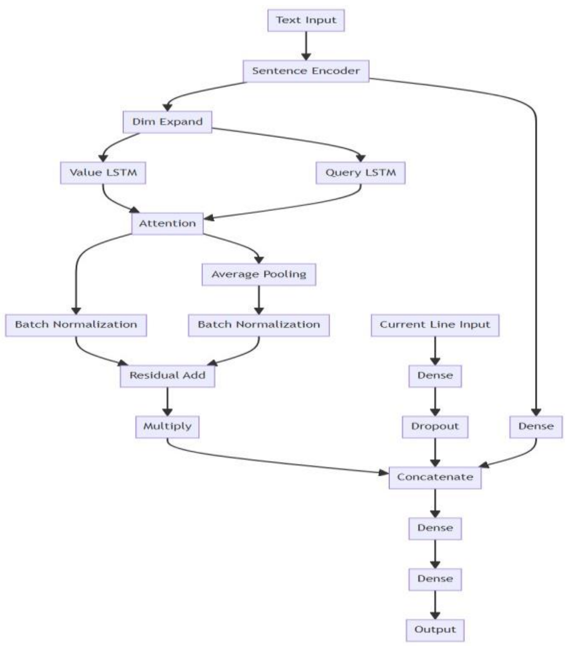

Figure 3: Self-attention with universal sentence encoder architecture

D. Training procedure

Hyperparameter tuning: Each model went through extensive hyperparameter tuning to find the optimal configurations for parameters like batch size, learning rate, number of epochs, and architecture-specific settings. A combination of grid search and random search techniques was used to explore the various hyperparameter combinations and find the most effective setup.

Fine-tuning pre-trained models: For models that incorporated pre-trained components like BERT and the Universal Sentence Encoder (USE), additional fine-tuning was conducted using the PubMed 200k RCT dataset [1]. Fine-tuning involved further training these models on the dataset, with adjustments made to the learning rate to avoid overfitting. This step was crucial for adapting the models to the domain-specific biomedical texts [3].

Data augmentation: To enhance model generalization and robustness, techniques such as shuffling sentences and replacing words with synonyms were applied. These augmentation methods introduced variability into the training set, which helped to enhance the models' ability to handle diverse and unseen data.

Training environment: All models were trained using TensorFlow and Keras frameworks within a GPU-enabled environment to speed up the training process. The use of GPUs was essential for efficiently handling large datasets and the complexity of deep learning models.

E. Evaluation metrics

Accuracy: The accuracy metric measured the ratio of correctly classified sentences to the total number of sentences. It served as a basic indicator of overall model performance.

Precision, recall, and F1-score: To access how effectively the models classified each of the five categories, precision, recall, and F1-score were calculated as depicted in figure 4 and figure 5. Precision measures the ratio of true positive predictions to the total predicted positives, while recall reflects the model's ability to identify all relevant instances. The F1-score, a balanced metric, combines both precision and recall, offering a more complete evaluation of performance.

Confusion matrix: Confusion matrices were employed to visualize the classification results as depicted in figure 6. These matrices detailed the relationship between the actual and predicted labels, offering insights into the model's strengths and potential areas for improvement across each category.

Comparison of models: Performance for each model was compared based on the above metrics, with a particular emphasis on accuracy and F1-score. These comparisons helped to identify the relative strengths and weaknesses of different architectures, aiding in the final analysis and conclusions of the study.

IV. RESULT

This section will present the model results and analyze their performance across multiple metrics, including accuracy, precision, recall, and F1-score. The results are examined with an emphasis on the implications of the different model architectures and the influence of various embedding strategies on classification performance.

A. Model performance comparison

The models were compared primarily using F1 scores, as this metric balances precision and recall, offering a comprehensive performance evaluation. This approach is particularly useful for imbalanced datasets such as PubMed 200k RCT [1]. Figure 4 shows the F1 scores for each model.

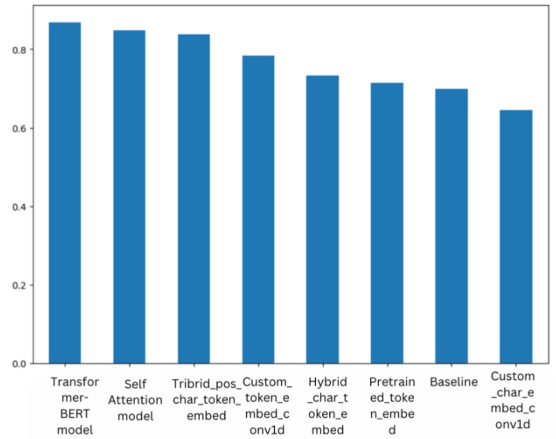

Figure 4: A comparison of F1 scores among the various models utilized in the study.

The Transformer+BERT model with positional encoding (Model 6) achieved the highest F1 score of 0.87 as shown in table 1, demonstrating its strong capability in classifying biomedical text sequences. The Self-Attention model combined with the Universal Sentence Encoder (Model 7) also performed well, with an F1 score close to Model 6, indicating its effectiveness in capturing dependencies across text [9]. Another model of interest was the Tribrid model, which incorporated token, character, and positional embeddings, further illustrating the benefits of combining multiple types of embeddings for improved performance. In contrast, the baseline model and models based solely on character embeddings showed comparatively lower F1 scores, emphasizing the value of more complex architectures and the inclusion of positional information in improving classification accuracy.

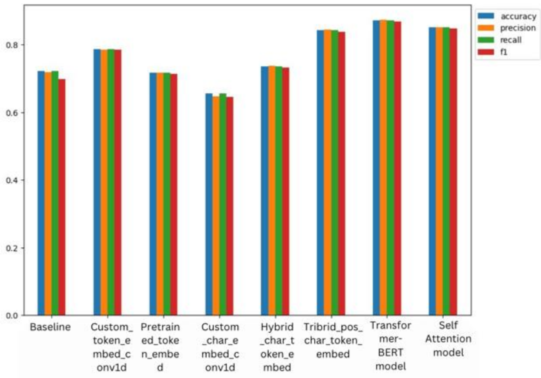

Figure 5: Comparison of accuracy, precision, recall, and f1 scores across different models used in the study

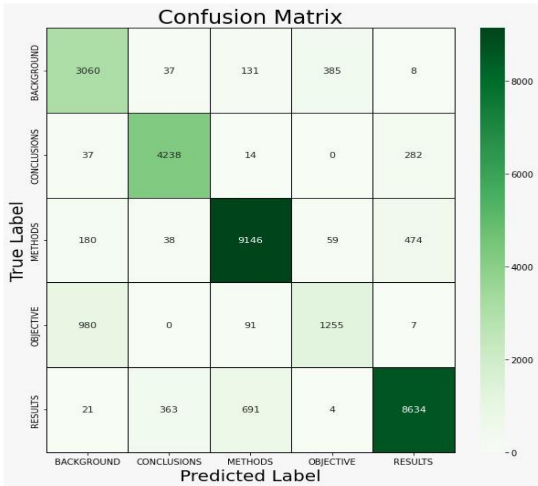

Figure 6: Confusion matrix

B. Analysis of embedding strategies

The experiments highlighted the critical role of embedding strategies in model performance. The Transformer+BERT model (Model 6), which integrated positional encoding, demonstrated superior results by leveraging sentence structure in biomedical abstracts, where sentence order is important. This approach allowed the model to capture contextual relationships effectively, enhancing its ability to classify biomedical text.

Character embeddings also contributed positively, especially in handling domain-specific terminology and abbreviations common in biomedical literature.

The Tribrid model, which combined token, character, and positional embeddings, further illustrated the advantage of using multiple embedding strategies, leading to more robust representations of complex text.

Table 1: F1-score for model 6

CLASS	F1-SCORE
BACKGROUND	0.77
CONCLUSIONS	0.89
METHOD	0.91
OBJECTIVE	0.63
RESULTS	0.90
ACCURACY	0.87

In contrast, simpler models, such as those relying only on token embeddings without additional context or positional information, struggled to achieve competitive performance. This underscores the necessity of more sophisticated embedding strategies to improve model accuracy in biomedical text classification.

C. Discussion

The findings of this study offer key insights for biomedical text classification. The success of the BERT with Positional Encoding model (Model 6) highlights the importance of incorporating both contextual and structural information in improving classification accuracy in complex text domains like biomedicine. Additionally, the strong performance of the Self-Attention model with Universal Sentence Encoder underscores the value of attention mechanisms in capturing long-range dependencies, which is crucial for biomedical texts where sentence structure and relationships are significant. One key issue is the difficulty in handling sentences with ambiguous or overlapping content.

V. CONCLUSION

This study examined the application of deep learning models for biomedical text classification, focusing on combining different embedding strategies to improve model performance. We explored the use of token, character, and positional embeddings across neural network frameworks, including CNNs, RNNs, Transformer models like BERT, and self-attention mechanisms.

A. Key findings

- The BERT with Positional Encoding model (Model 6) achieved the highest F1 score of 0.87 as shown in table 1 [6]. The model's ability to combine BERT's contextual understanding with positional information was crucial for accurately classifying biomedical text sequences.
- The Self-Attention with Universal Sentence Encoder model (Model 7) also performed well, effectively capturing long-range dependencies within the text, showing the utility of attention mechanisms paired with strong sentence encoders [7].
- The use of character-level embeddings across several models proved important for handling domain-specific terms common in biomedical texts. This highlights the benefit of incorporating fine-grained text representations alongside broader context-based embeddings.

B. Implications for biomedical text classification

The results have important implications for building more accurate biomedical text classification systems. The success of transformer-based models like BERT with Positional Encoding indicates that further exploration and refinement of such architectures could enhance classification performance, particularly in domains where text structure and context are critical [1].

The study also underscores the importance of embedding strategies that capture various aspects of text, including semantic meaning at the token level, detailed structure at the character level, and overall sequence information through positional encoding.

C. Limitations

Despite the strong model performance, several challenges remain. The models' dependence on large datasets and significant computational resources might restrict their applicability in environments with limited resources. Furthermore, the complexity of the models raises interpretability concerns, especially in biomedical applications where transparency is vital.

D. Final remarks

This study contributes to the field of biomedical text classification by showing how advanced deep learning models and embedding strategies can enhance performance. The insights gained can support researchers and developers in building more effective systems for analyzing biomedical literature, helping to inform better decisions in clinical and research contexts.

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